

UNIT – 4
ASSESSMENT – 3
PROBLEMS

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EXPERIMENT 1 — DECISION TREE CLASSIFICATION

1. PROBLEM

Title

Decision Tree Classification

Problem Statement

Implement and evaluate a Decision Tree classifier using the **Iris dataset**.

The program should:

1. Load the Iris dataset.
2. Pre-process the data and check for missing values.
3. Display the first five rows and data types.
4. Split the dataset into training and testing sets.
5. Build a Decision Tree using the **Gini Index**.
6. Print the tree structure.
7. Identify the root node and justify its selection.
8. Evaluate the model using:
 - Accuracy
 - Confusion Matrix
 - Classification Report

9. Display at least two misclassified instances.

EXPERIMENT 2 — REINFORCEMENT LEARNING: Q-LEARNING

1. PROBLEM

Title

Q-Learning in a 4×4 Grid World

Problem Statement

Implement a Q-Learning algorithm for a **4×4 Grid World**.

The environment contains:

- States
- Four actions: Up, Down, Left and Right
- Goal state
- Obstacle
- Reward system

The reward structure is:

Goal → +10

Normal → -1

Obstacle → -5

The agent must learn an optimal path to reach the goal.

The Q-Learning algorithm must:

1. Initialize the Q-table with zeros.
2. Define α , γ and ϵ .
3. Run for at least 100 episodes.
4. Record Q-values at Episodes 1, 50 and 100.

5. Display how the Q-table evolves.
6. Extract the final policy.
7. Plot cumulative reward per episode.
8. Analyze convergence.